

RESEARCH ARTICLE

Polymer
COMPOSITES

WILEY

Study on the anchoring performance and failure mechanism of basalt/glass hybrid fiber reinforced plastic anchors in coupled environment

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Funding information

Fundamental Research Funds for the Central Universities, Grant/Award Number: 3072023CFJ0206; National Natural Science Foundation of China, Grant/Award Number: 51708092; Natural Science Foundation of Heilongjiang Province, Grant/Award Number: YQ2023E031; Postdoctoral Research Foundation of China, Grant/Award Number: 2018M631894

[Correction added on 12 October 2023, after first online publication: The order of affiliations 1 and 2 have been swapped in the affiliation list and the affiliation numbers in author by-line have been updated accordingly.]

Abstract

To investigate the real anchorage performance of basalt and glass hybrid fiber reinforced plastic (FRP) anchors in external corrosive environments. In this paper, a combination of experimental studies and finite element analysis was used. By carrying out pull-out tests of basalt and glass hybrid FRP anchors under the synergistic effect of freeze–thaw cycles and alkaline environment, the effects of factors such as hole diameter, anchorage length, and mortar strength on their anchorage performance and durability were investigated. Critical corrosion time was introduced to determine the degree of corrosion of the anchor rods. After that, it combined with numerical simulation to reveal the degradation mechanism of the anchorage performance of hybrid anchors. The results showed that the best anchorage performance of basalt/glass hybrid FRP anchors was achieved when the hole diameter was 24 mm, the anchorage length was 175 mm, and the mortar strength was 45.5 MPa. The ultimate pull-out load was up to 122.16 kN. When the time did not exceed the critical corrosion time, the ultimate load-carrying capacity of basalt and glass hybrid FRP anchors increased by 9.3% compared to the uncorroded environment. Beyond the critical corrosion time, the ultimate load-carrying capacity was reduced by 23.3% and the anchorage performance was degraded. In addition, the contact interface between the anchor and the mortar is the weak area of the anchoring structure, which should be focused on in the later design of the actual project.

Highlights

- Basalt/glass hybrid FRP anchors have good corrosion resistance.
- The anchoring performance of hybrid FRP anchors under the coupled environment is studied.
- Hole diameter and anchoring length have more influence on the anchor performance.
- The concept of critical corrosion time is put forward.
- The failure mechanism and coupling environment influence are revealed by FES.

KEYWORDS

composites, failure, fibers, structure–property relations

1 | INTRODUCTION

The main function of soil slope is to prevent soil erosion. It stabilizes the soil and prevents water or wind from washing it away. This is important to protect vegetation, maintain farmland, protect water sources, and prevent erosion of rivers and coastlines. In addition, soil slopes can provide stability to geological structures. In the slope area, soil slope can resist the occurrence of geological disasters such as landslides and slope collapse. However, instability, landslides, or slumps can occur on solid slopes due to factors such as environment, slope shape, soil properties, and human damage. Therefore, the soil slope has been reinforced over a long period with anchors. The main function of an anchor is to increase the anti-sliding ability of the soil slope and reduce the deformation and settlement of the soil slope. To achieve the key effect of improving the overall stability of the soil slope. The soil slope has been anchored by steel anchors for a long time.¹ However, the serious corrosion problems of steel anchors in recent years have aroused widespread concern in the industry. Corrosion of steel anchors makes the long-term durability of anchors and the safety of building structures worrying.² The research shows that passive anti-corrosion measures are difficult to fundamentally solve the problem of steel anchors rust. It will also increase the cost of the project, which is not worth the loss.³ Therefore, new materials have been sought to replace steel anchors. At present, FRP has received extensive attention.^{4–8} This is mainly because the biggest advantage of FRP anchors compared with steel anchors is that they show ultra-high corrosion resistance in acid, alkali, salt, and freeze–thaw environments. This is the steel bolt does not have. FRP is made of glass, carbon, aramid, basalt, and matrix material by winding, forming, pultrusion, and other forming processes.^{9,10} The main advantages of single-fiber FRP composites are lightweight.^{11,12} And FRP composites have high strength, corrosion resistance, and can replace traditional steel anchors.^{13,14}

Domestic and foreign scholars have carried out detailed research on the anchoring of FRP anchors.^{15–20} The material itself is first studied experimentally. It is found that the diameter of the anchor, the rib spacing, the rib height, and the anchorage angle of the anchor have a great influence on the overall anchorage performance. Hao et al.²¹ conducted an experimental study on the diameter, rib spacing, and rib height of the anchor. The results show that the bond strength between FRP rib and

concrete is very high. However, the bond strength decreases with the increase in the diameter of the FRP rib. The failure mode of the specimen is a shear failure between the FRP rib and the concrete. And there is light wear on the surface of the ribs. Different rib parameters, such as rib spacing and rib height, have a significant influence on the bond performance between FRP rib and concrete. The optimum rib spacing of FRP rib is 1 times of rib diameter, and the optimum rib height is 6% of the rib diameter. Ozbakkaloglu et al.²² explored the Angle of the anchor and found that the pulling performance of the anchor decreases with the increase of the anchor Angle. Some scholars have analyzed the failure mechanism of FRP anchor.^{23–25} Two main factors cause the failure of FRP anchors. The first is fiber fracture, the second is fiber interface debonding, especially hybrid FRP anchors are the most prone to the latter failure mode.

At present, besides the material itself, three factors affect the anchoring performance of FRP anchors: the hole diameter, the type of grouting, and the anchoring length. Zhao et al.²⁶ found that with the increase in the strength grade of cement mortar, the bond strength between FRP bars and cement mortar gradually increases. However, with the increase of the diameter of FRP bars, the bond strength between FRP bars and cement mortar gradually decreases. Under the same conditions, the bonding strength of FRP bars with sand is higher than that of FRP bars with smooth surfaces. Saeed et al.²⁷ investigated the bonding length and curing time of grout based on the consideration of mortar strength and FRP bar diameter. It is found that the strength and stiffness of anchors are proportional to the anchoring length. The cross-sectional area ratio between the FRP bar and the anchor hole affects the stiffness and anchoring ability of the anchor. The recommended minimum bonding length is 32 times that of a 10 mm FRP bar and 30 times that of a 13 mm bar. In addition, through the comparison of curing time, it is also concluded that the tensile strength of the FRP anchor under 4-day curing is 6% lower than that under 4-week curing. Sun²⁸ and Soares et al.²⁹ studied the effect of seawater on the mechanical properties of grout. The former replaces traditional cement by preparing potassium magnesium phosphate cement. Compared with ordinary Portland cement, the bond strength of basalt-reinforced epoxy polymer (BFRP) bars and glass-reinforced epoxy polymer (GFRP) bars increased about 51.06% and 24.42%, respectively. This can indicate that potassium magnesium phosphate cement mortar tends to replace Portland cement mortar. In addition, when seawater is used instead

of freshwater as the mixed water. The bond strength between potassium magnesium phosphate cement and FRP bars is increased by 13.75%. This is mainly because the weak alkaline environment of seawater is conducive to the reaction of potassium magnesium phosphate with fly ash. And then promote the hardening of potassium magnesium phosphate cement to form a denser structure. To improve the bond strength. The latter test also proves that the use of seawater has no negative effect on the mechanical properties and bonding behavior of cement. Anchorage length is the main factor affecting the anchorage performance of FRP anchors. Feng et al.³⁰ drew some conclusions by comparing the tests on the anchoring performance of basalt fiber anchors and steel anchors. Based on the field pulling test of basalt fiber anchor and steel anchor in loess stratum, the failure mode and anchoring performance difference of the two materials are preliminarily studied. The ultimate pulling force of the basalt fiber anchor with the same anchoring length is similar to that of the steel anchor. All of them increase with the increase of anchorage length. The first interface shear resistance of steel anchors is incomparable to that of basalt fiber anchors. Under the test conditions, the reference value of bonding strength at the first interface of the basalt fiber anchor is from 0.4 to 0.7 MPa. The interface bonding strength between the steel anchor and grout is from 1.7 to 2.6 MPa.

However, the above test results only consider the geological factors of the site. They do not consider the influence of the environment on the two kinds of anchors. The influence of the environment on anchoring performance is particularly important under actual working conditions. Ahmed et al.³¹ investigated the anchoring performance of FRP anchors in dry, wet, freeze-thaw, and other environments. When the buried depth length is five times the diameter of the FRP anchor, the FRP anchor will slip and fail. It is recommended to increase the buried depth to 15 times the diameter of the FRP anchor. In the freeze-thaw cycle environment, Piao³² and Zhang et al.³³ investigated the anchoring performance of anchors. The study variables of the former are 25, 50, and 75 days of freeze-thaw period test block load test. The results show that the freeze-thaw cycle will lead to the deterioration of the bond between epoxy resin mortar and concrete. The loss of bearing capacity after 25, 50, and 75 days is 38.24%, 42.65%, and 52.94%, respectively. The freeze-thaw cycle accelerates the deterioration of material properties of bonded materials and surrounding rocks. The load transfer to the interior is accelerated. So that the stress difference between the points on the FRP anchor gradually decreases. The latter uses the indoor model test to obtain the changes in anchor displacement, anchorage force, anchor stress, surrounding rock

stress, and shear stress at the intersection of anchor and mortar under different freeze-thaw cycles. The results show that the elastic modulus and strength of the mortar are reduced by the freeze-thaw cycle. The mortar failure at the loading end is advanced, which speeds up the transfer of load to the depth of the anchor. The stress of anchor and shear stress at the interface between anchor and mortar increases. The ultimate load of the anchor decreases and the deformation increases under the action of the freeze-thaw cycle. With the increase of the freeze-thaw period, the inflection point of the load-displacement curve and the horizontal section of the steel bar slip curve appear in advance.

In previous studies, domestic and foreign scholars have only investigated the effect of a single environment on anchorage performance. This is not applicable to the real environment. Generally, in the real environment will be affected by at least two or more complex environments on the anchorage performance. Especially in areas where freeze-thaw environments exist. The influence of three factors, namely, hole diameter, anchorage length, and mortar strength, on anchorage performance has not been investigated in depth. In this paper, experimental study, theoretical analysis, and finite element analysis are used to investigate the anchorage performance of basalt/glass hybrid FRP anchors in depth. The premise is the synergistic effect of the freeze-thaw cycle and alkaline environment. The optimum hole diameter, the optimum anchorage length, and the optimum mortar strength are determined. Finally, the concept of critical corrosion time is derived from the test results. The deterioration mechanism of the anchoring structure is then analyzed. This provides a high theoretical reference value for the subsequent anchorage application of basalt/glass hybrid FRP anchors in practical engineering.

2 | MATERIALS AND EXPERIMENTAL PROCEDURES

2.1 | Raw materials

In this paper, the cross-section appearance and anchoring form of basalt/glass hybrid FRP anchors are shown in Figure 1A. As shown in the figure, the hybrid anchor consists of two parts. The shell structure is a basalt-reinforced epoxy polymer, and the inner core structure is a glass-reinforced epoxy polymer. The length of basalt/glass hybrid FRP anchors is 400 mm, and both ends are anchored 80 mm. The basic properties of basalt fiber, glass fiber, and epoxy resin used in this test are shown in Table 1. According to the previous work, the design of

FIGURE 1 Anchoring structure profile and concrete strength test: (A) Diagram of basalt/glass composite fiber rod, (B) standard part test, (C) anchor structure.

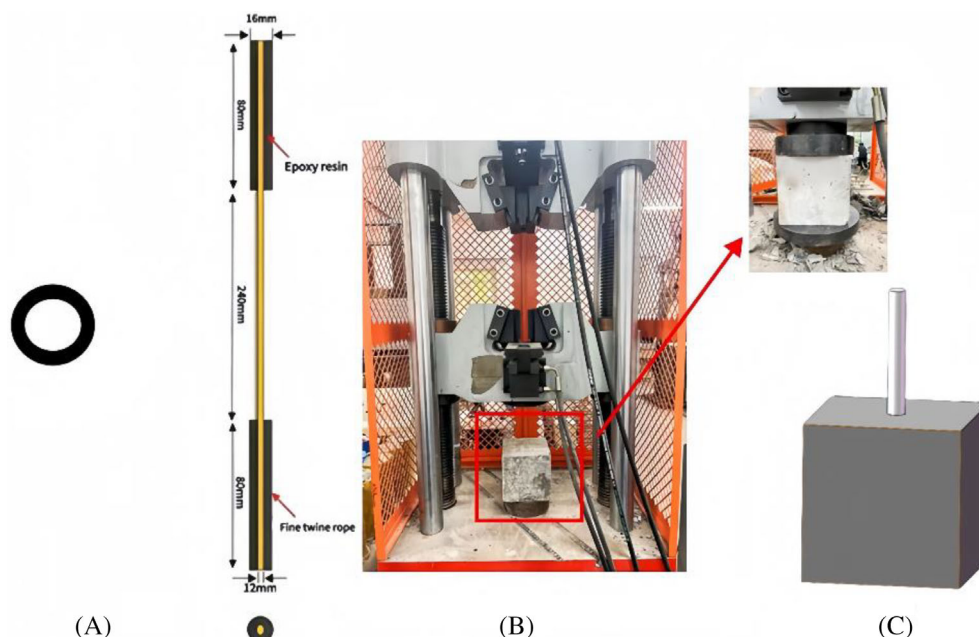


TABLE 1 Mechanical properties of basalt fibers, glass fibers, and epoxy resin.

Materials	Tensile strength (MPa)	Tensile modulus (GPa)	Rupture strain (%)
Basalt fiber	3500	120	1.97
Glass fiber	1350	73	3.41
Epoxy resin	52.4	3.51	≥4

this structure can significantly improve the elongation of hybrid anchors.^{34,35} This is because fibers with high elongation can delay the destruction of low-ductility fibers. In addition, the outer basalt fiber has good corrosion resistance and insulation properties.^{36–38} It can prevent the electrochemical corrosion of the glass fiber core. The mechanical properties of the feedstock and the production process of the hybrid FRP anchors have been mentioned in previous studies.^{39,40} In addition, the total fiber volume fraction of hybrid FRP anchors accounts for about 70%, and the epoxy resin glue accounts for 30%. The volume fraction ratio of basalt fiber to glass fiber is set at 3:7. To ensure that the hybrid anchors have a certain elastic modulus and anchoring performance. Concrete blocks are used to simulate slopes for this test. Raw materials and preparations for making concrete are shown in Table 2. The cube concrete size is 200 mm × 200 mm × 200 mm, as shown in Figure 1B. The strength of the standard cube concrete for 28 days of curing is tested. The compressive strength can reach 43.6 MPa, which meets the requirements of this test. In addition, the combined structure of

the basalt/glass hybrid FRP anchor and concrete block is shown in Figure 1C.

2.2 | Test scheme

2.2.1 | Coupled environment

The simulated environment in this paper is the environment of the freeze–thaw cycle and alkali solution soaking simultaneously. Therefore, the above two environments are called coupled environments in this paper. To simulate the temperature change in the real environment, the author consults the morning and evening temperature difference report of Harbin City in March. It could be found that the temperature difference between morning and evening tends to be −8 to 10°C. Therefore, to simulate the real environment. When specimens are doing the freeze–thaw cycle test, the temperature range of the low and high temperature test chamber should be adjusted to −10 to 10°C. Each freeze–thaw cycle is 6 h with 0, 40, and 100 cycles. The temperature change rate of the test chamber is fast. The temperature deviation is not more than 2.0°C, and the temperature adjustment range is from −120 to 150°C. It is controlled digitally by special software. The temperature can simulate the real temperature according to linear, sinusoidal, or mixed changes, as shown in Figure 2. The PH of the alkaline solution should be around 13.2, and the composition of the alkaline solution is shown in Table 3. The Alkali soaking time is 25 and 80 days.

The test scheme of the influence of the coupled environment on the anchoring performance of basalt/glass

TABLE 2 Mix proportion of concretes (units: $\text{kg}\cdot\text{m}^{-3}$).

Type	Cement	Water	Sand	Gravel	Water reducing agent	Defoamer
C40 concrete	450	180	611	1185	1.80	0.072



FIGURE 2 High and low-temperature test chamber.

TABLE 3 Composition and pH of basic solutions.

Ca(OH) ₂	NaOH	KOH	pH
2.2 g/L	2.5 g/L	19.4 g/L	13.2

hybrid FRP anchors is shown in the following Table 4. The changing parameters are the number of freeze–thaw cycles and the soaking days of the alkali solution. It can be seen from Table 4 that the cycles of freezing and thawing are 0, 40, and 100 times respectively. The soaking time of the alkali solution is 0, 25, and 80 days, respectively. The test consists of three groups with three specimens in each group.

BG in the first item of the specimen group number stands for basalt/glass hybrid FRP anchors; 12 stands for 12 mm single rib; Items 1, 2, and 3 in the third item are the numbers of the test pieces in the experimental group; In the fourth item, (0), (1), and (2), respectively represent freeze–thaw cycles of 0, 40, and 100 times, and the soaking days of alkali solution are 0, 25, and 80 days.

2.2.2 | Hole diameter, anchoring length, and mortar strength

In this test, the anchoring length of basalt/glass hybrid FRP anchors, the central hole diameter of the concrete block, and the strength of the mortar are designed by Design Expert software, as shown in Table 5. It could be seen that the anchoring length ranges from 150 to

200 mm, the hole diameter ranges from 18 to 30 mm, and the mortar strength ranges from 41.5 to 45.5 MPa. Because the coefficient of variation is very small, each group of anchors is processed three. Then the calculated data is averaged.

Test scheme under the influence of hole diameter

The groups of hole diameter tests are shown in Table 6. The change parameter is the reserved hole diameter of a 12 mm single basalt/glass hybrid FRP anchor. The hole diameter is divided into 18, 24, and 30 mm. The test consists of three groups with three specimens in each group.

Test scheme under the influence of anchoring length

The anchoring length test groups are shown in Table 7. Anchors of the same diameter has different load-bearing capacity and distribution under different anchoring lengths. Taking the anchoring length as a variable, the bearing capacity load distribution of different anchoring lengths is tested. The anchoring length is divided into 150, 175, and 200 mm. The test consists of three groups with three specimens in each group.

Test scheme under the influence of mortar strength

The test groups are shown in Table 8. The load-bearing capacity of anchors with the same diameter is different under different mortar strengths. Taking mortar strength as a variable, the influence of different mortar strengths on the bearing capacity distribution of basalt/glass hybrid FRP anchors is tested. Mortar strength is divided into 41.5, 43.5, and 45.5 MPa three kinds. The test consists of three groups with three specimens in each group.

2.2.3 | Test equipment and process

To measure the change of stress inside the anchor during drawing, strain gauges are attached along the length direction of basalt/glass hybrid FRP anchors. The strain gauge is arranged longitudinally at intervals of 20 mm along the anchoring length. First, polish the basalt/glass hybrid anchors' surface with an Angle grinder. After grinding, clean with alcohol, and then paste the strain gauge. To prevent damage to the strain gauge during agitation, the strain gauge paste is first fixed with quick

TABLE 4 Experimental design in the coupled environment.

Serial number	Longitudinal reinforcement (mm)	Hole diameter (mm)	Length (mm)	Number of freeze-thaw cycles	Alkaline soaking days
BG-12-1(0)	12	18	400	0	0
BG-12-2(0)	12	18	400	0	0
BG-12-3(0)	12	18	400	0	0
BG-12-1(1)	12	18	400	40	25
BG-12-2(1)	12	18	400	40	25
BG-12-3(1)	12	18	400	40	25
BG-12-1(2)	12	18	400	100	80
BG-12-2(2)	12	18	400	100	80
BG-12-3(2)	12	18	400	100	80

TABLE 5 Factors and levels of simulation test.

Serial number	Anchorage length (mm)	Hole diameter (mm)	Mortar strength (MPa)
1	150	24	41.5
2	175	24	43.5
3	150	18	43.5
4	150	24	45.5
5	150	30	43.5
6	175	30	41.5
7	200	30	43.5
8	200	24	45.5
9	175	30	45.5
10	200	24	41.5
11	175	18	45.5
12	175	18	41.5
13	200	18	43.5

adhesive and then coated with AB adhesive for protection. The drawing test is carried out on the MTS800 electro-hydraulic servo loading system. The method of uniform loading is adopted, and the loading rate is 2 mm/min. DH3815N static strain test system is used to measure the strain at each test point during the drawing test. The special drawing test device is shown in Figure 3. The top of the tool is connected to the upper fixture, and the bottom is suspended. The tension end of the anchor is anchored in advance. During the test, the tension end of the anchor rod is connected to the clamp fixed to the lower beam of the MTS through the round hole at the bottom of the tool. The upper connection and the lower connection make the tension of the

TABLE 6 Experimental design under the influence of hole diameter.

Serial number	Longitudinal reinforcement (mm)	Hole diameter (mm)	Length (mm)
BG-12-18-1	12	18	400
BG-12-18-2	12	18	400
BG-12-18-3	12	18	400
BG-12-24-1	12	24	400
BG-12-24-2	12	24	400
BG-12-24-3	12	24	400
BG-12-30-1	12	30	400
BG-12-30-2	12	30	400
BG-12-30-3	12	30	400

Note: BG in the first item of the specimen group number stands for basalt/glass hybrid FRP anchors. 12 stands for 12 mm single rib. In the third item, 18, 24, and 30 represent reserved holes. 1, 2, and 3 are the number of the specimen in the test group.

anchor along its axis. The center hole diameter at the bottom of the tool is 30 mm, which is about 15 mm larger than the reserved hole diameter of the anchor. An electronic displacement meter is installed at the free end and the loading end of the anchor respectively. The slip quantity of basalt/glass hybrid FRP anchors during loading is collected by the DH3815N static strain test system.

To obtain the stress distribution of the anchor section of anchor and the change of stress distribution during loading. Strain gauges are uniformly arranged on one side of basalt/glass hybrid FRP anchors. The axial force distribution at the corresponding points of basalt/glass hybrid FRP anchors can be obtained by calculating

TABLE 7 Experimental design under the influence of anchoring length.

Serial number	Longitudinal reinforcement (mm)	Hole diameter (mm)	Length (mm)
BG-12-150-1	12	24	400
BG-12-150-2	12	24	400
BG-12-150-3	12	24	400
BG-12-175-1	12	24	400
BG-12-175-2	12	24	400
BG-12-175-3	12	24	400
BG-12-200-1	12	24	400
BG-12-200-2	12	24	400
BG-12-200-3	12	24	400

Note: BG in the first item of the specimen group number stands for basalt/glass hybrid FRP anchors. 12 stands for 12 mm single rib. In the third item, 150, 175, and 200 represent the anchoring length of basalt/glass hybrid FRP anchors. 1, 2, and 3 are the number of the specimen in the test group.

TABLE 8 Experimental design under the influence of mortar strength.

Serial number	Longitudinal reinforcement (mm)	Hole diameter (mm)	Length (mm)
BG-12-41.5-1	12	24	400
BG-12-41.5-2	12	24	400
BG-12-41.5-3	12	24	400
BG-12-43.5-1	12	24	400
BG-12-43.5-2	12	24	400
BG-12-43.5-3	12	24	400
BG-12-45.5-1	12	24	400
BG-12-45.5-2	12	24	400
BG-12-45.5-3	12	24	400

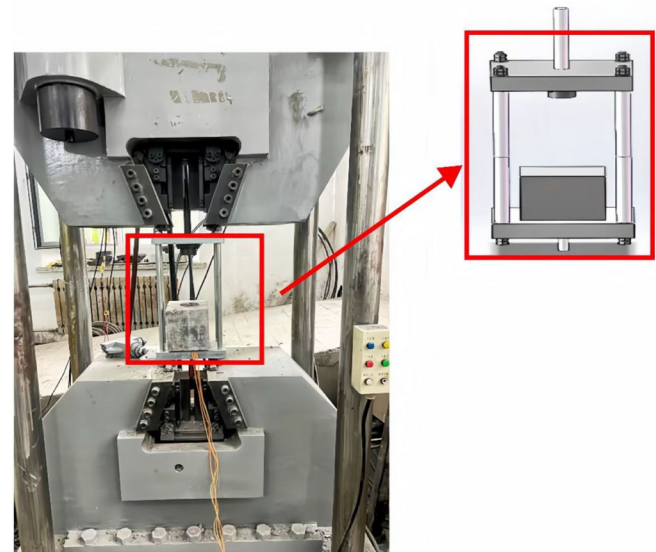
Note: BG in the first item of the specimen group number stands for basalt/glass hybrid FRP anchors. 12 represents 12 mm single rib. In the third item, 41.5, 43.5, and 45.5 represent mortar strength. 1, 2, and 3 are the number of the specimen in the test group.

the strain values measured by the strain gauges at each point on the basalt/glass hybrid anchor. The formula is shown below.

$$N_x = \mu \epsilon_x \cdot EA, \quad (1)$$

where $\mu \epsilon_x$ is the surface strain of basalt/glass hybrid FRP anchors; E is the elastic modulus of basalt/glass hybrid FRP anchors; A is the cross-sectional area of basalt/glass hybrid FRP anchors.

During the test, the displacement of the free end of the anchor is measured by a dial indicator. However,

**FIGURE 3** Apparatus of the pull-out test.

slippage between anchors and mortar and between mortar and concrete blocks is more practical. The free end displacement of the anchor is composed of five parts: elastic elongation of basalt/glass hybrid FRP anchors, slippage between anchor and mortar, slippage between mortar layer and concrete block, elastic deformation of mortar layer, and elastic deformation of concrete block. Since most basalt/glass hybrid FRP anchors' pulling occurs mixed interface damage, the slip between anchor and mortar, mortar and concrete can be considered comprehensively. The expression of slip quantity is:

$$\delta_s = \delta_e - \delta_b - * \delta_m - * \delta_c, \quad (2)$$

where δ_s is the slip amount of basalt/glass hybrid FRP anchors and mortar; δ_e is the measurement value of the anchor end displacement; $* \delta_m$, $* \delta_c$ are the deformation of anchoring mortar and concrete block respectively; δ_b is the elastic elongation of basalt/glass hybrid FRP anchors free section.

3 | RESULTS AND DISCUSSION

3.1 | Influencing factor on the anchoring performance of basalt/glass hybrid FRP anchors

3.1.1 | Effect of coupled environment

Figure 4A–C show the load–displacement curves of basalt/glass hybrid FRP anchored structures under different cycle times and different alkali soaking time. Figure 4A shows the results of the control group. From

Figure 4B, it could be seen that a small section of the curve in the initial loading stage is relatively flat, mainly caused by uneven force on the bottom of the concrete block. From the comparison between Figure 4A,B, it could be seen that the average maximum load of the latter is 95.68 kN, higher than that of the former's 87.5 kN. It is about 9.3% higher than the average maximum load of the former. The average maximum displacement of the latter is 10.92 mm, which is slightly higher than the former's 10.05 mm. It is about 8.7% higher than the average maximum displacement of the former. The main reason for this interesting phenomenon is that the immersion time and the number of cycles are relatively short, resulting in no deep damage to basalt/glass hybrid FRP anchored structures. The corrosion effect is not enough to reduce the strength of concrete blocks and mortar. In addition, basalt/glass hybrid FRP anchors have excellent corrosion resistance. After experiencing short-term corrosion, the surface will appear some corrosion phenomenon. This corrosion phenomenon, on the contrary, increases the friction on the surface of basalt/glass hybrid FRP anchors, which makes basalt/glass hybrid FRP anchors and mortar have a greater gripping force. Therefore, the overall anchoring structure presents a larger average load and relatively large slip amount. In this paper, the time when the above phenomena occur is called the Critical corrosion time point. The above phenomenon can also be explained from a micro perspective. The main components in cement are silicates (tricalcium silicates and dicalcium silicates). When an alkali solution reacts with the silicates in cement, the alkali hydroxide causes alkali aggregation. During alkali aggregation, alkali ions (sodium hydroxide, potassium hydroxide, etc.) migrate and aggregate in the concrete in an ionic form. This aggregation process leads to the formation of pores and an increase in void ratio, which reduces the strength and durability of the concrete.

In Figure 4A,C, it could be found that the overall performance of the anchor structure tends to decline when the critical corrosion time point is exceeded. The average maximum load of the latter is about 23.3% lower than that of the former. The average maximum displacement of the latter is 22.4% lower than that of the former. The main reason for this phenomenon is that corrosion in the coupled environment has penetrated deep into the concrete, mortar, and basalt/glass hybrid FRP anchors. With the passage of time, the internal voids of concrete and mortar gradually increase in the coupled environment. This causes the alkali solution to enter the interior of both more quickly and accelerates the deterioration of concrete and mortar properties. The mechanical properties of basalt/glass hybrid FRP anchors also decrease gradually. This is because, in the coupled environment, there is debonding

between the two fibers in basalt/glass hybrid FRP anchors and the accompanying fiber corrosion. The anchoring performance of basalt/glass hybrid FRP anchors decreases sharply under the combined action of these two factors.

From the comparison of Figure 4, it could be found that there is indeed a critical corrosion time point in the coupled environment. When the corrosion critical time point is not exceeded, the overall anchoring performance increases instead of decreasing. When the corrosion critical time point is exceeded, the overall performance will be greatly reduced. As can be seen from the comparison of Figure 4B,C. The difference in the number of freeze-thaw cycles and soaking time of alkali solution between the two is 60 times and 55 days, respectively. The difference between the average maximum load and the average maximum displacement is 24.69 kN and 2.71 mm. Therefore, it could be concluded that with the passage of time, the anchoring force of basalt/glass hybrid FRP anchoring structures increases first and then decreases. The slope of the increasing section is lower than that of the descending section when the critical anchorage time is exceeded. When the critical corrosion time is exceeded, cracks develop on the surface of the cement paste grout. As the corrosion time progresses, the cracks expand from the exterior to the interior. This allows water and alkali solutions to enter the interior of the anchored structure through the cracks. This results in a decrease in bond strength at the first interface. Currently, it is possible to repair damaged cement grout with self-healing granules. According to previous studies, the repair rate of self-healing particles can reach 70%–90%.⁴¹ In addition microorganisms (*Bacillus subtilis* and soil bacteria) both contribute to the partial healing of cracks.⁴²

3.1.2 | Effect of hole diameter

For basalt/glass hybrid FRP anchors of the same length, the anchoring failure under different hole diameters is manifested as sliding failure between the anchor solid and the rock mass (basalt/glass hybrid FRP anchors are bonded to the mortar as a whole). Also known as second interface destruction. It could be seen from the drawing load-displacement curve that the hole diameter has a great influence on the ultimate drawing load of basalt/glass hybrid FRP anchors. The bearing capacity of basalt/glass hybrid FRP anchors decreases immediately after sliding, and the final residual anchorage strength is only 15% ~ 35% of the maximum anchorage strength. When the hole diameter is larger, the residual friction will also increase. Because the friction contact surface is also larger.

According to the comparison of slope changes in the early stage and the late stage shown in Figure 5A–C. It

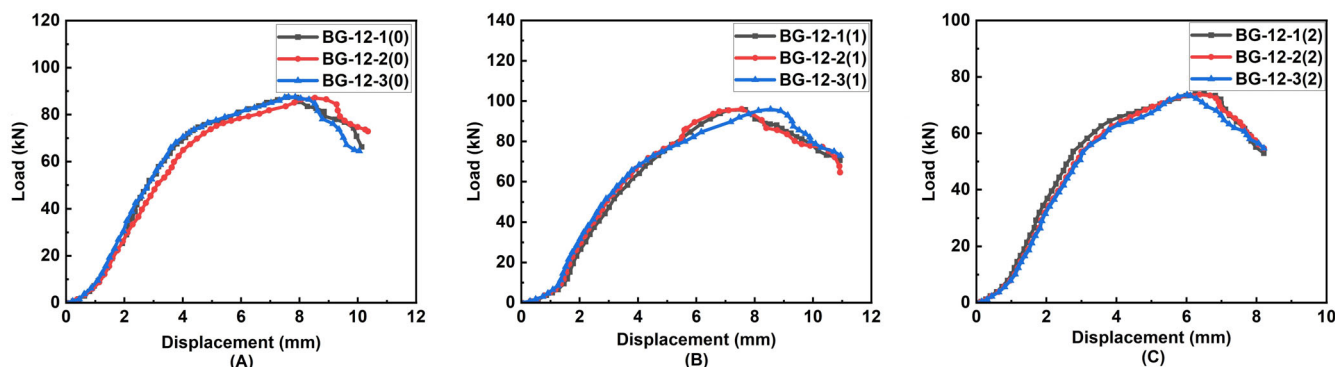


FIGURE 4 Load displacement curve under coupled environment: (A) Uncoupled environment, (B) Freeze–thaw cycle 40 times, alkaline solution soaked for 25 days, (C) Freeze–thaw cycle 100 times, alkaline solution soaked for 80 days.

could be seen that basalt/glass hybrid FRP anchors do not slip in the early stage. The load always increases in direct proportion to the tension displacement. However, the slope of basalt/glass hybrid FRP anchors has decreased significantly in the later stage, and the anchoring section has begun to slip. When the hole diameter is 18 mm, the ultimate average load is 84.7 kN and the maximum average slippage is 10.14 mm. When the hole diameter is 24 mm, the ultimate average load is 95.52 kN and the maximum average slippage is 12.95 mm. When the hole diameter is 30 mm, the ultimate average load is 71.89 kN and the maximum average slippage is 10.07 mm. Compared with the above three figures, it could be seen that the larger the hole diameter, the faster the slippage of the anchor under the same drawing force. The slippage quantity increases first and then decreases. Similarly, the ultimate drawing strength of basalt/glass hybrid FRP anchors increases first and then decreases with the increase of the hole diameter. This is because the grout is too thick and the elastic modulus of the grout is lower. The overall deformation of the anchor and grout is more likely to occur, and the sliding failure is more likely to occur.

As shown in Figure 6A, through comparison, it could be seen that the ultimate pulling forces of basalt/glass hybrid FRP anchors with different hole diameters are successively 24 mm > 18 mm > 30 mm. The ultimate drawing displacement is 30 mm < 18 mm < 24 mm. When the hole diameter is 18 mm, the mortar thickness between basalt/glass hybrid FRP anchors and rock mass is only 2 mm, resulting in a small limit pulling force. When the diameter of the borehole is 30 mm, the small elastic modulus of grout leads to large deformation under load. Interlayer shear failure of grout may even occur. This results in a 32.9% lower ultimate pulling force than 24 mm. Therefore, in the actual construction, the diameter of the opening hole should be controlled and not be too large. The diameter of the opening hole is best matched with the diameter of the anchor, and

the thickness of the grout material is suitable for about 6 ~ 8 mm.

It can be seen from Figure 6A that the maximum drawing load under different hole diameters is 95.52 kN and the minimum is 71.89 kN. In this paper, the pull-out stiffness (the ratio of pull-out load to pull-out displacement) is calculated with a load of 65 kN at a three-hole diameter. When the drawing load is 65 kN, the corresponding drawing displacements are 3.45, 4.15, and 5.79 mm respectively. When the drawing load is 25 kN, the corresponding drawing displacements are 1.75, 2.12, and 2.67 mm, respectively. When the drawing load is 10 kN, the corresponding drawing displacements are 1.45, 1.6, and 1.63 mm, respectively. As shown in Figure 6B. It could be seen that with the continuous reduction of drawing load, the drawing stiffness under three kinds of hole diameters tends to be a straight line with a small slope. The slope increases with the increase of the drawing load. From the initial straight line slowly becomes a polygonal line. Therefore, it could be concluded that when the drawing load reaches 10.46% ~ 26.17% of the ultimate drawing load, the influence of hole diameters on the drawing stiffness is very small and could be ignored. Beyond this range, the drawing stiffness is inversely proportional to hole diameters. In addition, the thickness of the grout will lead to the reduction of stiffness, and this reduction is proportional to the thickness of the grout. Therefore, the shear stiffness reduction caused by grout thickness must be taken into account in the calculation of the basalt/glass hybrid FRP anchor's drawing ultimate bearing capacity.

3.1.3 | Effect of anchoring length

Anchoring length is a particularly important factor affecting the anchoring performance of basalt/glass hybrid FRP anchors. According to previous experimental

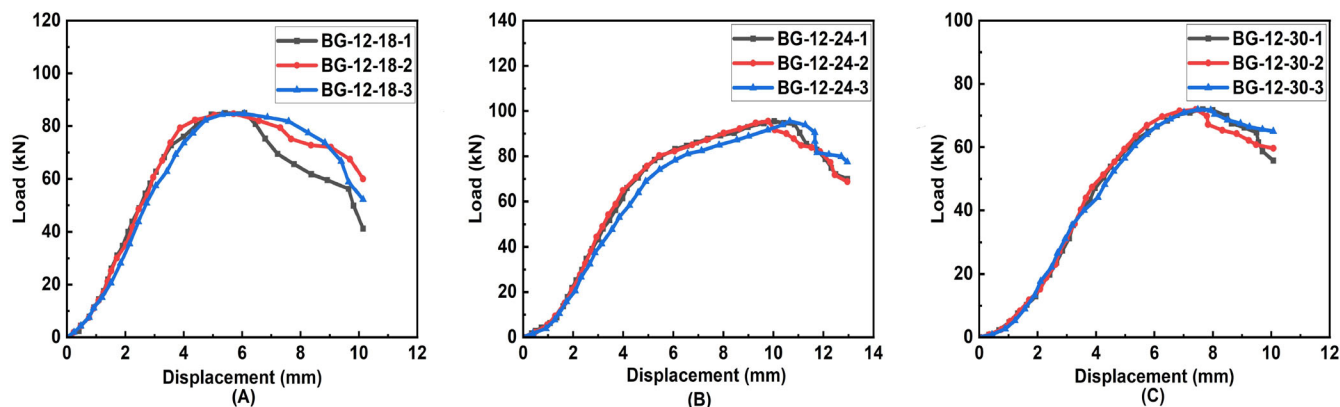


FIGURE 5 Load displacement curve under the influence of hole diameter: (A) 18 mm, (B) 24 mm, (C) 30 mm.

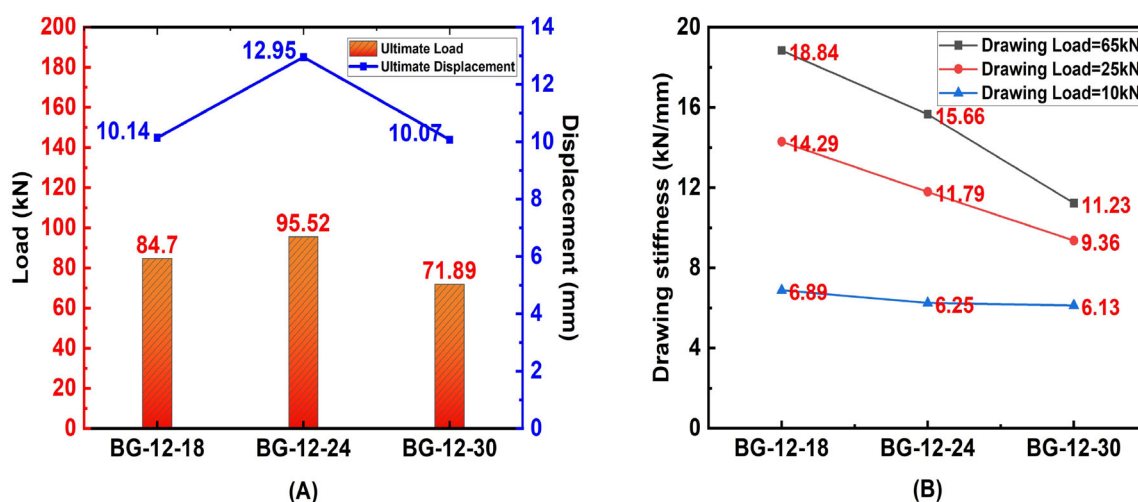


FIGURE 6 Effect of hole diameter on anchoring performance and drawing stiffness: (A) anchoring performance, (B) drawing stiffness.

studies, three types of anchoring failure modes are obtained: (1) the first interface failure (sliding failure due to insufficient bonding force between anchor and grout), (2) the second interface failure (sliding failure due to insufficient bonding force between anchor solid and rock mass), and (3) the anchor is broken.^{43,44} These three failure modes are closely related to the anchoring length of the anchor. By comparing Figure 7A–C, it could be seen that the greater the anchoring length, the greater the ultimate pulling force, and the smaller the ultimate pulling displacement of basalt/glass hybrid FRP anchors. When the anchoring length is 150 mm, the ultimate drawing load is 85.6 kN. When the anchoring length is 200 mm, the ultimate drawing load is 108.21 kN. The difference between the two limit drawing loads is 22.61 kN, and the latter is about 26.41% higher than the former.

By comparing Figure 7B,C, it could be found that both the anchoring failure modes are the anchoring failure caused by the basalt/glass hybrid FRP anchor's

fracture. The difference in ultimate drawing load between the two is only 1.26 kN. This is because the anchoring length of basalt/glass hybrid FRP anchors has reached the critical anchoring length.⁴⁵ Scholars at home and abroad have also conducted in-depth studies on the critical anchorage length. The difference in anchorage length will directly affect the difference of damage mode of the anchored structure. So the concept of critical anchorage length is given. So that in the future research can be well defined which damage mode will occur in the anchored structure.^{46,47} When the anchor is damaged, it can be seen that the load drops suddenly, and the anchor has lost its bearing capacity. Therefore, it could be concluded that the ultimate pulling strength of basalt/glass hybrid FRP anchors increases with the increase of anchoring length. However, when the anchoring length of basalt/glass hybrid FRP anchors exceeds 175 mm in this test, the increase of ultimate pulling load can be almost negligible. Therefore, basalt/glass hybrid FRP

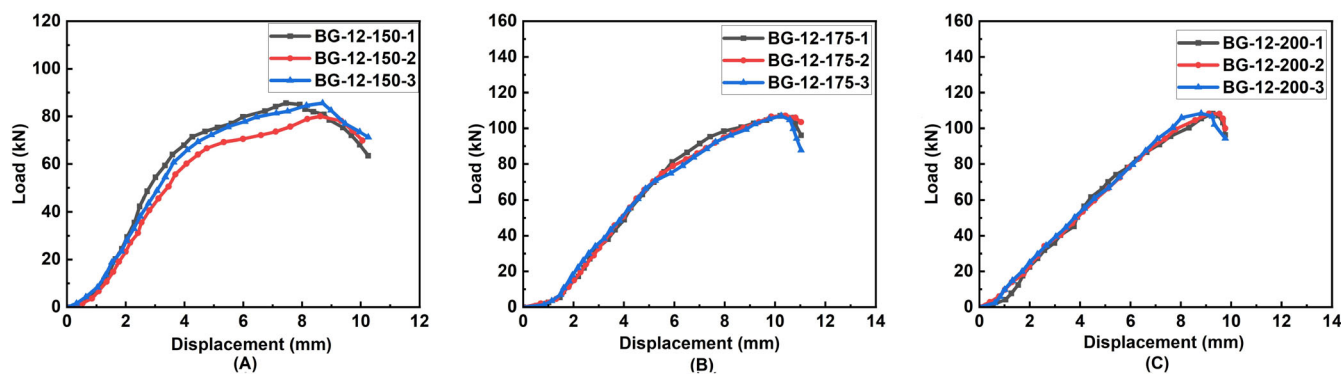


FIGURE 7 Load displacement curve under the influence of anchoring length: (A) 150 mm, (B) 175 mm, (C) 200 mm.

anchors can fully play the tensile performance of the rod body itself when the anchoring length is 175 mm.

It could be seen from Figure 8A–C of axial force distribution. After the tension load is applied to the anchor, the axial force on basalt/glass hybrid FRP anchors shows that the maximum axial force of the anchor appears at the junction of the anchored section and the free section. And it becomes smaller as the anchoring length increases. The load of basalt/glass hybrid FRP anchors gradually deepens along the whole length with the increase of loading force. Comparing Figure 8A,C, it can be seen that with the increase of loading force, the maximum stress depth difference between the two is 30 mm. This can also indicate that the external load received beyond the critical anchoring length will not continue to be transferred to the deep. However, after the length of the anchor reaches 175 mm, the anchoring length continues to increase, and the loading depth does not increase significantly. Through the above analysis, it could be seen that there is a critical value in the bonding length of basalt/glass hybrid FRP anchors. When the bonding length is less than this value, The specimens are predominantly damaged at the first or second interface. When the bonding length is not less than this value, basalt/glass hybrid FRP anchors are subject to fracture damage. It could be concluded that when the compressive strength of mortar is 43.5 MPa and the strength of concrete is 41 MPa, the critical anchoring length of basalt/glass hybrid FRP anchors in this test is 175 mm. It is about 14.5 times the cross-section diameter of basalt/glass hybrid FRP anchors. Therefore, the critical anchoring length of basalt/glass hybrid FRP anchors is about 14–15 times the section diameter.

3.1.4 | Effect of mortar strength

Compared with the hole diameter and anchoring length, it could be found that the increase in mortar strength will

also affect the anchoring performance of basalt/glass hybrid FRP anchors. But the improvement is not very significant. It can be seen from Figure 9 that when the mortar strength is 41.5, 43.5, and 45.5 MPa, the corresponding ultimate anchoring load is 94.32, 94.92, and 96.76 kN, respectively. The maximum growth rate is 2.58%. The maximum growth rates of anchorage load affected by anchoring length and hole diameter were 26.5% and 32.86%, respectively. Therefore, it could be concluded that the anchoring length and the hole diameter are the key factors affecting the anchoring performance. When these two factors play to the best level, the strength of the mortar can be considered to improve. To further improve the overall anchoring performance. In future studies, grout with greater bonding strength with basalt/glass hybrid FRP anchors can be sought to improve the lateral restraint ability and average bonding strength of anchor solids. For example, polymer-modified mortars, geopolymer mortars, and polyethylene terephthalate-modified mortars can be used to replace conventional cement mortars.^{48–50} This results in improved bond strength.

3.2 | Analysis of optimal anchorage performance of basalt/glass hybrid FRP anchors

The experiment is conducted according to the response surface design methodology. Influencing factors (Anchorage length, hole diameter, and mortar strength) are taken as independent variables. The anchoring capacity and slip amount of basalt/glass hybrid FRP anchors are used as evaluation indexes. The test results are shown in Table 9.

According to the data samples in Table 9, the quadratic polynomial regression model of anchorage bearing capacity and slip amount of basalt/glass hybrid FRP anchors are obtained by Design-Expert 8.0.6 software.

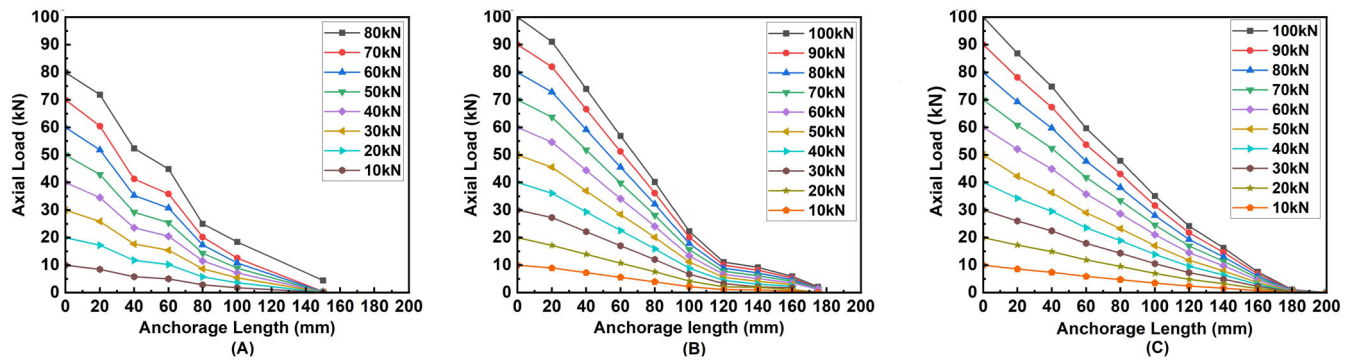


FIGURE 8 Axial force distribution curve influenced by anchorage length: (A) 150 mm, (B) 175 mm, (C) 200 mm.

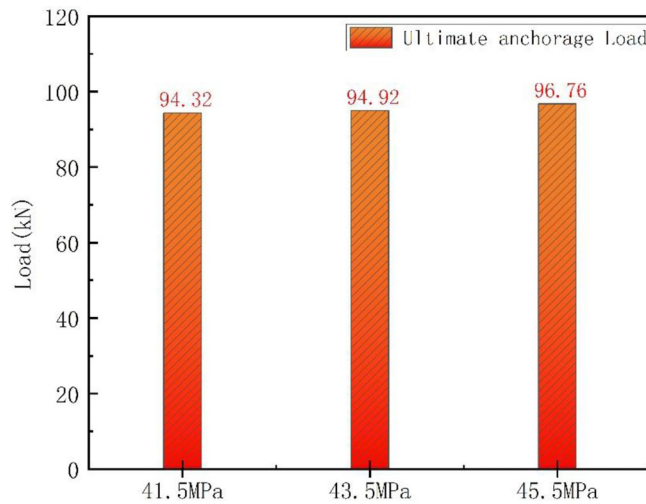


FIGURE 9 Influence of mortar strength on anchoring performance.

$$Y_1 = 121.47 + 13.69A - 5.33B + 0.6C + 0.98AB + 0.29AC + 0.44BC - 11.36A^2 - 19.63B^2 - 0.37C^2, \quad (3)$$

$$Y_2 = 14.1 - 0.28A - 0.22B - 0.075C - 0.15AB - 0.005AC - 0.015BC - 1.01A^2 - 3.27B^2 - 0.018C^2, \quad (4)$$

where Y_1 is the anchoring capacity (kN), Y_2 is the slip amount of basalt/glass hybrid FRP anchors (mm), A is the anchorage length (mm), B is the hole diameter (mm), C is the mortar strength (MPa).

According to the results of regression model analysis, Design-Expert 8.0.6 software is used to draw the 3D response surface diagram of the interaction effects of various factors. The response surface of anchorage length and hole diameter to anchoring bearing capacity is shown in Figure 10A. When the anchorage length increases from 150 to 200 mm and the hole diameter increases from 18 to

30 mm, the anchoring force increases first and then decreases. And when the hole diameter is unchanged, it can be seen that with the increasing of anchoring length, the anchoring bearing capacity also increases gradually. When the anchoring length is between 150 and 175 mm, the overall anchoring bearing capacity increases more than that when the anchoring length is between 175 and 200 mm. It could be concluded from this that when the anchorage length is exceeded, the anchorage bearing capacity will continue to increase but the increase will not be too large. This is mainly because the fiber of basalt/glass hybrid FRP anchors has been broken in a large area or the fiber debonding has occurred. Thus basalt/glass hybrid FRP anchors can no longer provide greater drawing capacity. It can be seen from the figure that the critical point is when the hole diameter is 24 mm and the anchoring length is about 175 mm.

The response surface of anchorage length and hole diameter to basalt/glass hybrid FRP anchors' slip is shown in Figure 10B. When the anchoring length increases from 150 to 200 mm and the hole diameter increases from 18 to 30 mm, the slip amount still shows a trend of increasing first and then decreasing. From the above two pictures, it can be seen that the anchoring length and hole diameter are indeed the key factors affecting the entire anchoring structure. When the anchoring length reaches 200 mm and the hole diameter reaches 24 mm, the overall anchoring bearing capacity of the anchoring structure is maximum. It can reach about 126 kN. When the anchoring length reaches 175 mm and the hole diameter reaches 24 mm, the overall slip of the anchoring structure is the largest, which can reach about 13 mm. This is in good agreement with the experimental results. However, based on the test data in Section 3.1.3 and the theoretical support of predecessors, it has been concluded that when the anchor reaches the critical anchoring length, the anchor will fully exert its strength until the anchor is destroyed. Therefore, the effect of the member when the critical anchorage length is exceeded is the same as that when the critical anchorage length is reached. To avoid the waste of

TABLE 9 Design and results of the orthogonal test.

Serial number	Anchorage length (mm)	Hole diameter (mm)	Mortar strength (MPa)	Anchorage bearing capacity (kN)	Slippage (mm)
1	150	24	41.5	94.32	13.44
2	175	24	43.5	121.47	14.1
3	150	18	43.5	84.7	10.14
4	150	24	45.5	96.76	13.25
5	150	30	43.5	71.89	10.07
6	175	30	41.5	95.11	10.62
7	200	30	43.5	98.22	9.2
8	200	24	45.5	125.74	12.71
9	175	30	45.5	97.36	10.47
10	200	24	41.5	124.12	12.88
11	175	18	45.5	106.95	11.03
12	175	18	41.5	106.47	11.12
13	200	18	43.5	107.12	9.85

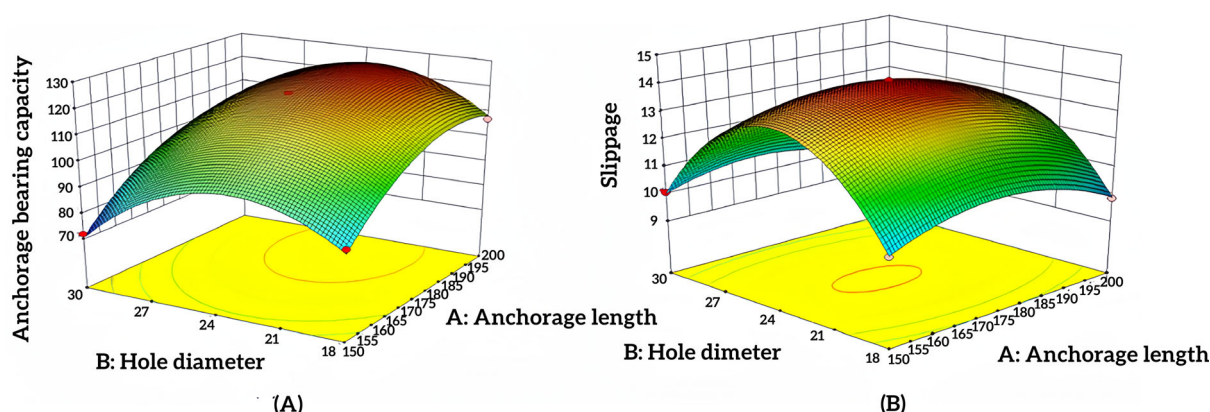


FIGURE 10 3D response diagram: (A) anchorage capacity, (B) slippage.

materials and time in practical engineering, it is generally recommended that the critical anchorage length is reached. The results show that when the hole diameter is 24 mm, the anchoring length is 175 mm, and the mortar strength is 45.5 MPa, the basalt/glass hybrid FRP anchors have the best anchoring performance, and the anchoring load is about 122.16 kN.

3.3 | Finite element simulation

3.3.1 | Anchoring performance mechanism analysis of basalt/glass hybrid FRP anchors

The anchoring performance of basalt/glass hybrid FRP anchors is simulated by ABAQUS software. The

mechanism of its anchoring failure is further revealed. Firstly, the cube concrete 3D model and basalt/glass hybrid FRP anchor's model are established. Concrete parameters are consistent with those in Section 2.1. The concrete model size is 200 mm × 200 mm × 200 mm. The basalt/glass hybrid FRP anchor's length is 400 mm. The x and y planes are built to cut the basalt/glass hybrid FRP anchor's model with a length of about 150 mm. The two models arrange the operation and retain the imprint. The next step is to mesh and remove the excess concrete nodes in the parts of the basalt/glass hybrid FRP anchors. In the same way, the superfluous nodes of basalt/glass hybrid FRP anchors are removed from the concrete parts. Assembling the two parts and adding batch connector springs in the anchorage length segment. As shown in Figure 11A.

FIGURE 11 Damage simulation of anchoring structure model: (A) establishment of anchoring structure model, (B) anchorage damage analysis.

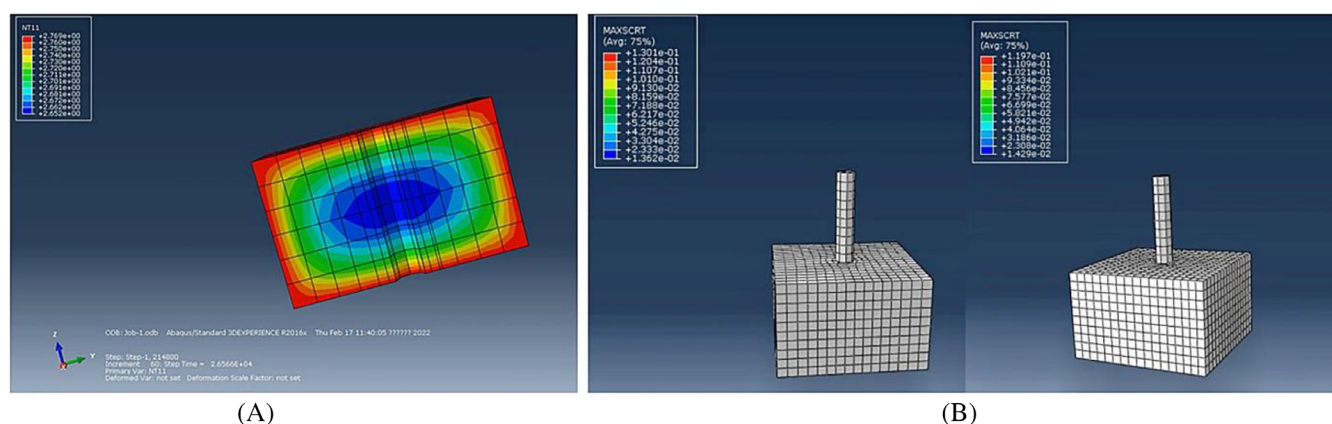
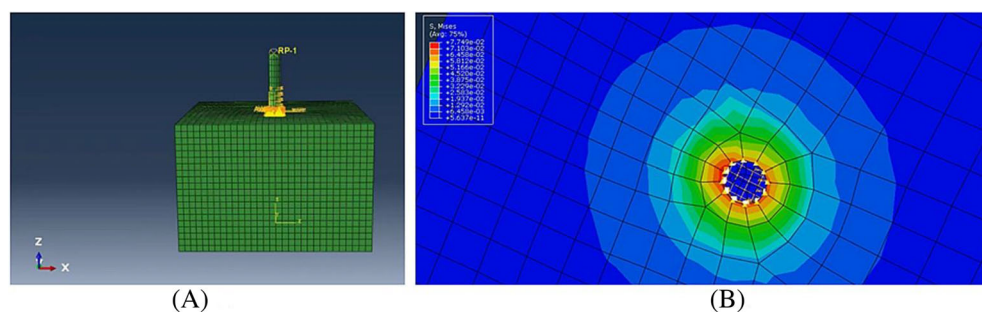


FIGURE 12 Damage simulation diagram of anchoring structure model in the coupled environment: (A) model damage analysis, (B) model damage comparison.

A stretching simulation is then performed, as shown in Figure 11B. It can be seen from the figure that the stress is greatest at the contact interface between the rod and the grout. Therefore, it could be seen that the interface is the weak area of the anchor structure. And the damage occurs first at the interface. Therefore, the strength and thickness of grout are the key factors to effectively resist external loads. When the external stress is greater than the tensile strength of the anchor but less than the bond strength of the grout, the whole anchor structure will be pulled apart. This is the most ideal result in practical engineering. Therefore, the selection of grouting material type and thickness is very important for the entire anchoring structure. The grout with the best combination effect with basalt/glass hybrid FRP anchors should be further explored in future research. Thus, the anchoring performance of basalt/glass hybrid FRP anchoring structures can be improved indirectly. In addition, it could be seen that the stress away from the interface becomes less and less. This is mainly due to the high strength of the grout and concrete, resulting in the stress being absorbed by both in turn. Therefore, it could be concluded that the ability to absorb stress depends on the strength grade of the grout and concrete.

3.3.2 | Mechanism analysis of basalt/glass hybrid FRP anchoring performance under the coupled freeze–thaw cycle and alkaline environment

First of all, the three-dimensional model of cube concrete is established with dimensions of $200\text{ mm} \times 200\text{ mm} \times 200\text{ mm}$. A circular hole with a radius of 8 mm is created in the middle. A bonding unit is established on the inner surface of the hole. After that, the basalt/glass hybrid FRP anchor's model is established with a diameter of 12 mm and a length of 400 mm . The x and y planes are established to cut the basalt/glass hybrid FRP anchor's model with a cutting length of 150 mm . To simulate a real alkaline environment, the PH of the alkaline solution is set to 13.2. The freeze–thaw cycle temperature range is -10 to 10°C . The simulation established 50 freeze–thaw cycles, varying every 600 seconds. Then choose heat transfer. Figure 12A shows the model state under the combined action of freeze–thaw cycle and alkaline environment. The anchor is assembled with the concrete model, and then the grid is divided.

As seen from Figure 12B, there are obvious differences between the anchoring performance in the coupled

environment and the uncoupled environment at the same time. The damage degree of the former is $1.301\text{e-}01$, while the damage degree of the latter is $1.197\text{e-}01$. It could be seen that the coupled environment does affect the performance of anchor solids. The damage degree will gradually increase with the increase of time. The damage mechanism is mainly due to the sharp decline in the performance of concrete, grout, and basalt/glass hybrid FRP anchors under the action of the coupled environment (compressive strength of concrete, bond strength of grout, interface bond strength between fibers in basalt/glass hybrid FRP anchors). In the simulation, it could be seen that the last failure mode of the structure is the first interface (the interface between the anchor and the grout) or the second interface (the interface between the grout and the concrete) at $0 \sim 50$ cycles. Because basalt fiber has good corrosion resistance, it is more affected by the environment for concrete and grout. It can also be seen that the bond strength of the interface of the two fibers in the basalt/glass hybrid FRP anchors remains good after 50 cycles. This also shows that basalt fibers and glass fibers have a better ability to prevent water infiltration. However, with the increase of coupled environment time, basalt/glass hybrid FRP anchors may be destroyed before concrete and grout. Further research is needed to calculate the damage rates of basalt/glass hybrid FRP anchors, concrete, and grout under the influence of a coupled environment through tests and simulations.

4 | CONCLUSIONS

In this paper, experimental studies and finite element analysis are used. Hole diameter, anchorage length, and mortar strength were used as variables. The optimal hole diameter, optimal anchorage length, and optimal mortar strength are determined by response surface analysis. The degradation mechanism of anchorage performance and durability of basalt/glass hybrid FRP anchors in coupled environments is also analyzed. The following conclusions could be drawn based on the research results and discussions.

The combined effect of freeze–thaw cycles and alkaline solution indeed results in a decrease in overall anchorage performance. When subjected to 100 freeze–thaw cycles and 80 days of alkaline solution immersion, the ultimate bearing capacity decreases by 23.3% compared to the non-coupled environment. The weakening of the ultimate bearing capacity intensifies as time progresses. Additionally, in the coupled environment, a critical corrosion time emerges over time. When the duration of exposure to the coupled environment does not exceed the critical

corrosion time, the overall anchorage performance does not decrease but rather improves. The experimental results show that when subjected to 40 freeze–thaw cycles and 25 days of alkaline solution immersion, the ultimate bearing capacity increases by 9.3% compared to the non-coupled environment. Under the same drawing force, the larger the hole diameter, the faster the slip of basalt/glass hybrid FRP anchors. The slip quantity increases first and then decreases. Similarly, the ultimate bearing capacity of basalt/glass hybrid FRP anchors also increases first and then decreases with the increase of the hole diameter. This is because the grout is too thick and the elastic modulus of the grout is low. The deformation of basalt/glass hybrid FRP anchors and grout is more likely to occur, resulting in sliding failure. Therefore, the thickness of the grout material is suitable for about $6 \sim 8$ mm. When the tensile load reaches $10.46\% \sim 26.17\%$ of the ultimate tensile load, the effect of hole diameter on the tensile stiffness is small and negligible. In addition, this test again confirms the existence of critical anchorage length. In this test, the critical anchoring length is 175 mm, which is about 14.5 times the section diameter of basalt/glass hybrid fiber anchors. Therefore, the critical anchoring length of basalt/glass hybrid FRP anchors is recommended to take 14 times or 15 times the section diameter of the bar.

It could be found that the strength of mortar also affects the anchoring performance of basalt/glass hybrid FRP anchors. But the degree of improvement is less. The test results show that when the mortar strength increases from 41.5 to 45.5 MPa, the ultimate bearing capacity only increases by 2.58%. Through the Design-Expert software, the influence of three factors on the overall anchoring structure under the coupled environment is analyzed. The analysis results are in good agreement with the experimental results. The results show that when the hole diameter is 24 mm, the anchoring length is 175 mm, and the mortar strength is 45.5 MPa, the anchoring capacity of basalt/glass hybrid FRP anchors reaches the maximum, about 122.16 kN. This could be used as a reference in practical engineering.

The failure mechanism of basalt/glass composite FRP bolt is further revealed by finite element simulation. The simulation results show that the contact interface between the anchor and slurry is the weak area of the anchoring structure. The damage occurs first at the interface. Therefore, the type of slurry and the surface form of the bolt should be deeply explored in future research to meet the needs of practical engineering. Finally, it is verified by simulation that the damage degree of the whole anchoring structure increases with the increase of coupling environment time.

ACKNOWLEDGMENTS

This work is financially supported by the Natural Science Foundation of Heilongjiang Province with grant no. YQ2023E031, the Fundamental Research Funds for the Central Universities with grant no. 3072023CFJ0206, the National Natural Science Foundation of China (NSFC) with grant no. 51708092, and the Postdoctoral Research Foundation of China with grant no. 2018M631894.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sets generated during the current study are available from the corresponding author on reasonable request.

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REFERENCES

- Zhu H, Xiang Q, Luo B, Du Y, Li M. Evaluation of failure risk for prestressed anchor cables based on the AHP-ideal point method: an engineering application. *Eng Fail Anal.* 2022;138:106293.
- Cheng L. Present status and development of ground anchorages. *China Civil Eng J.* 2001;10:7-12. +34.
- Han J, Zhang M, Zhang H, et al. Rebar bolt optimization for large deformational roadways with application case. 2019; 48(4):727-734.
- Saxena M, Pappu A, Sharma A, Haque R, Wankhede S. *Composite materials from natural resources: recent trends and future potentials*. IntechOpen; 2011.
- Pleimann LG. Strength, Modulus of Elasticity, and Bond of Deformed FRP Rods, *Advanced Composites Materials in Civil Engineering Structures*, ASCE, 1991, pp. 99–110.
- Chaallal O, Benmokrane B. Glass-Fiber Reinforcing Rod: Characterization and Application to Concrete Structures and Grouted Anchors, *Materials: Performance and Prevention of Deficiencies and Failures*, ASCE, 1992, pp. 606–617.
- Adam MA, Said M, Mahmoud AA, Shanour AS. Analytical and experimental flexural behavior of concrete beams reinforced with glass fiber reinforced polymers bars. *Construct Building Mater.* 2015;84:354-366.
- Ali NM, Wang X, Wu Z. Integrated performance of FRP tendons with fiber hybridization. *J Compos Construct.* 2014;18(3):A4013007.
- Ren Y, Wang H, Guan Z, Yang K. Evaluation of the properties and applications of FRP bars and anchors: a review. *Rev Adv Mater Sci.* 2023;62(1):20220287.
- Liu Z, Wang H, Yang L, Du J. Research on mechanical properties and durability of flax/glass fiber bio-hybrid FRP composites laminates. *Compos Struct.* 2022;290:115566.
- Wang H, Yang K, Guan Z, Gao S. Orthogonal study on mechanical and tension-tension fatigue properties of flax/glass fiber hybrid FRP composites. *Fibers Polym.* 2023;24(6):2173-2193.
- Zhang M, Guan Z, Ren Y, Wang H. Multi-scale study on the mechanical properties of pultruded GFRP laminates. *Fibers Polym.* 2023;24(6):2133-2146.
- Luo J, Peng Z, Wang X, Yan Z, Lin J, Liu L. Mechanical behavior of novel basalt fiber-reinforced polymer (BFRP) rock bolt. *Compos Mater Sci Eng.* 2022;12:79-86.
- Zhao J, Mei K, Wu J. Long-term mechanical properties of FRP tendon-anchor systems—a review. *Construct Build Mater.* 2020;230:117017.
- Malvar LJ. Tensile and bond properties of GFRP reinforcing bars. *Mater J.* 1995;92(3):276-285.
- Cosenza E, Manfredi G, Realforzo R. 20 Analytical Modelling of Bond between FRP Reinforcing Bars and Concrete, *Non-metallic (FRP) Reinforcement for Concrete Structures: Proceedings of the Second International RILEM Symposium*, CRC Press. 1995 p. 164.
- Rossetti VA, Galeota D, Giammatteo M. Local bond stress-slip relationships of glass fibre reinforced plastic bars embedded in concrete. *Mater Struct.* 1995;28:340-344.
- Cosenza E, Manfredi G, Realforzo R. Behavior and modeling of bond of FRP rebars to concrete. *J Compos Construct.* 1997; 1(2):40-51.
- Okelo R, Yuan RL. Bond strength of fiber reinforced polymer rebars in normal strength concrete. *J Compos Construct.* 2005; 9(3):203-213.
- Gao D, Zhu H, Xie J. The constitutive models for bond slip relation between FRP rebars and concrete. *Indus Construct.* 2003;33(7):41-43.
- Hao Q, Wang Y, Hou J, Ou J. Experimental study on bond behavior of GFRP Ribbed rebars. *Eng Mech.* 2008;10:158-165. +179.
- Ozbakkaloglu T, Saatcioglu M. Tensile behavior of FRP anchors in concrete. *J Compos Construct.* 2009;13(2):82-92.
- Li C, Guo R, Xian G, Li H. Innovative compound-type Anchor-rod system for a large-diameter pultruded carbon/glass hybrid rod for bridge cable. *Mater Struct.* 2020;53:1-15.
- Zhang H, Smith ST, Kim S. Optimisation of carbon and glass FRP anchor design. *Construct Build Mater.* 2012;32:1-12.
- Pan Y, Yan D. Study on the durability of GFRP bars and carbon/glass hybrid fiber reinforced polymer (HFRP) bars aged in alkaline solution. *Compos Struct.* 2021;261:113285.
- Zhao D, Wen S, Wang L, Zhang B, Yang L. Structural parameters and critical anchorage length of tunnel system bolts made of basalt fibre. *Construct Building Mater.* 2021;310:125081.
- Saeed YM, Al-Obaidi SM, Al-Hasany EG, Rad FN. Evaluation of a new bond-type anchorage system with expansive grout for a single FRP rod. *Construct Building Mater.* 2020;261:120004.
- Sun W, Zheng Y, Zhou L, Song J, Bai Y. A study of the bond behavior of FRP bars in MPC seawater concrete. *Adv Struct Eng.* 2021;24(6):1110-1123.
- Soares S, Freitas N, Pereira E, Nepomuceno E, Pereira E, Sena-Cruz J. Assessment of GFRP bond behaviour for the design of sustainable reinforced seawater concrete structures. *Construct Building Mater.* 2020;231:117277.
- Feng J, Wang Y, Zhang Y, Huang L, He C, Wu H. Experimental comparison of anchorage performance between basalt fiber and steel bars. *Rock Soil Mech.* 2019;40(11):4185-4193.
- Ahmed EA, El-Salakawy EF. Tensile capacity of GFRP post installed adhesive anchors in concrete. *J Compos Construct.* 2008;12(6):596-607.

32. Piao S, Zhang W, Wang Y, Deng L. Experimental study on the influence of freeze-thaw cycles on the anchoring performance of GFRP anchor. *Bull Geol Sci Technol*. 2022;41(6):301-307.
33. Zhang W, Xiang W, Jia H. Experimental study of mechanical properties of grouted rock under freeze-thaw cycles. *Chinese J Rock Mech Eng*. 2014;33(3):558-566.
34. You Y-J, Park Y-H, Kim H-Y, Park J-S. Hybrid effect on tensile properties of FRP rods with various material compositions. *Compos Struct*. 2007;80(1):117-122.
35. Guo R, Li C, Xian G. Water absorption and long-term thermal and mechanical properties of carbon/glass hybrid rod for bridge cable. *Eng Struct*. 2023;274:115176.
36. Wang Z, Zhao X, Xian G, et al. Long-term durability of basalt-and glass-fibre reinforced polymer (BFRP/GFRP) bars in seawater and sea sand concrete environment. *Construct Building Mater*. 2017;139:467-489.
37. Li Y, Yin S, Lu Y, Hu C. Experimental investigation of the mechanical properties of BFRP bars in coral concrete under high temperature and humidity. *Construct Building Mater*. 2020;259(30):120591.
38. Lu C, Ni M, Chu T, He L. Comparative investigation on tensile performance of FRP bars after exposure to water, seawater, and alkaline solutions. *J Mater Civil Eng*. 2020;32(7):04020170.
39. Guo R, Xian G, Li C, Huang X, Xin M. Effect of fiber hybridization types on the mechanical properties of carbon/glass fiber reinforced polymer composite rod. *Mech Adv Mater Struct*. 2022;29(27):6288-6300.
40. Guo R, Xian G, Li C, et al. Water uptake and interfacial shear strength of carbon/glass fiber hybrid composite rods under hygrothermal environments: effects of hybrid modes. *Polym Degrad Stab*. 2021;193:109723.
41. Zhang C, Liu R, Chen M, Li X, Zhu Z, Yan J. Effects of independently designed and prepared self-healing granules on self-healing efficiency for cement cracks. *Construct Building Mater*. 2022;347:128626.
42. Acarturk BC, Sandalci I, Hull NM, Bundur ZB, Burris LE. Calcium sulfoaluminate cement and supplementary cementitious materials-containing binders in self-healing systems. *Cement Concr Compos*. 2023;141:105115.
43. Wang L, Song Z, Yi J, Li J, Fu F, Qian K. Experimental studies on bond performance of BFRP bars reinforced coral aggregate concrete. *Intern J Concr Struct Mater*. 2019;13(1):1-10.
44. Hu J. *The Research and Analysis on Anchoring Performance of GFRP Bolt*. Central South University; 2012.
45. Wang H, Bai X, Zhang M, Yan N, Wang Y. Research Progress of basalt fiber reinforced polymer anchors in rock and soil anchoring. *Compos Mater Sci Eng*. 2020;8:113-122.
46. Huang M, Zhao M, Chen C. Influence of anchorage length on stress in bolt and its critical value calculation. *Rock Soil Mech*. 2018;39(11):4033-4041. +4062.
47. Zhao X, Sun K, Huang Q, et al. Research on the reasonable anchorage length and failure characteristics of resin anchorage system. *Construct Building Mater*. 2022;356:129309.
48. Rizaee S, Shrive N. Shear bond strength of adhered thin masonry veneer with traditional and polymer modified mortars. *Construct Building Mater*. 2023;379:131277.
49. Peng K, Zeng J, Huang B, Huang J, Zhuge Y, Dai J. Bond performance of FRP bars in plain and fiber-reinforced geopolymer under pull-out loading. *J Build Eng*. 2022;57:104893.
50. Lee M, Kim K, Chung C-W, Kim W, Jeong Y, Lee J. Mechanical characterization of recycled-PET fiber reinforced mortar composites treated with nano-SiO₂ and mixed with seawater. *Construct Build Mater*. 2023;392:131882.

How to cite this article: Ren Y, Wang H, Wu G, Guan Z, Yuan L. Study on the anchoring performance and failure mechanism of basalt/glass hybrid fiber reinforced plastic anchors in coupled environment. *Polym Compos*. 2024;45(1):946-962. doi:10.1002/pc.27828